

Problem Set 1 – Solution

Microeconomic Theory 3 – ECON 709, Fall 2026

This document provides a solution sketch for each answer. Sometimes, the suggested solution is of course not the only "right" answer, but you might have come up with different, equally correct answers.

1. (after Kreps, Problem 1.9.) We say that a binary relation $\succ$ on X is *negatively transitive* if, for any $x, y, z \in X$, $x \succ y$ implies that either $z \succ y$ or $x \succ z$ (or both).

Prove the following:

Suppose a binary relation $\succ$ is asymmetric and negatively transitive. Define $\succsim$ by $x \succsim y$ if not $y \succ x$, and define $\sim$ by $x \sim y$ if neither $x \succ y$ nor $y \succ x$. Then $\succsim$ is reflexive, complete, and transitive, and if we defined $\sim'$ and $\succ'$ from $\succsim$ as given in the lecture notes, then $\sim'$ would be the same as $\sim$, and $\succ'$ would be the same as $\succ$.

Solution:

First show that $\succsim$ is reflexive, complete, and transitive:

Take any $x, y \in X$ (not necessarily distinct). Either [not $x \succ y$] or [$x \succ y$ which implies not $y \succ x$ by asymmetry]. Hence either not $x \succ y$ or not $y \succ x$. Thus either $y \succsim x$ or $x \succsim y$. Hence, $\succsim$ is complete and reflexive.

Let $x \succsim y$ and $y \succsim z$ for some $x, y, z \in X$. Thus not $y \succ x$ and not $z \succ y$. Suppose by contradiction that not $x \succsim z$. Then $z \succ x$. Negative transitivity then implies that either $y \succ x$ or $z \succ y$ or both, a contradiction. Thus it must be that $x \succsim z$ and $\succsim$ is transitive.

Second show that $\sim'$ would be the same as $\sim$:

$$x \sim y \Leftrightarrow \text{not } x \succ y \text{ and not } y \succ x \Leftrightarrow y \succsim x \text{ and } x \succsim y \Leftrightarrow x \sim' y.$$

Third show that $\succ'$ would be the same as $\succ$:

$x \succ y \Rightarrow \text{not } y \succsim x$ (by def. of $\succsim$) and not $y \succ x$ (by asymmetry of $\succ$)
 $\Rightarrow x \succsim y$ (by def. of $\succsim$) and not $y \succsim x \Rightarrow x \succ' y$.

$x \succ' y \Rightarrow \text{not } y \succsim x \Rightarrow x \succ y$.

2. Kreps, Problem 1.6. Solution taken from his solution manual.

■ 1.6. (a) Suppose that $x \succeq y$ and $y \succeq z$ for $x = (x_1, \dots, x_k)$, $y = (y_1, \dots, y_k)$, and $z = (z_1, \dots, z_k)$. Then $x_i \geq y_i$ and $y_i \geq z_i$ for each i . Hence, by transitivity of $\geq$ for real numbers, $x_i \geq z_i$ for all i , and thus $x \succeq z$. This shows that $\succeq$ is transitive.

On the other hand, for $k = 2$, $x = (1, 2)$, and $y = (2, 1)$, neither $x \succeq y$ nor $y \succeq x$; $\succeq$ is not complete.

(b) $x \succ y$ if $x \succeq y$ and not $y \succeq x$, which is $x_i \geq y_i$ for all i , and not $y_i \geq x_i$ for all i , which is $x_i \geq y_i$ for all i , and $x_i > y_i$ for some i .

This is asymmetric: If $x \succ y$, then $x_i > y_i$ for some i . Thus neither $y \succeq x$ nor $y \succ x$ are possible.

But this is not negatively transitive: Take $x = (2, 2)$, $z = (1, 1)$, and $y = (3, 0)$. We have $x \succ z$, but neither $x \succ y$ nor $y \succ z$ is true.

(c) $x \sim y$ if $x \succeq y$ and $y \succeq x$, which is $x_i \geq y_i$ and $y_i \geq x_i$ for all i , which is $x_i = y_i$ for all i . Thus we have

$$x \sim y \text{ if and only if } x = y.$$

This is clearly reflexive, symmetric, and (trivially) transitive.

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3. (Rubinstein, Lecture 1, Problem 1. The solution is taken from Rubinstein's solution manual.) Let $\succsim$ be a rational preference relation on a set X . Define $I(x)$ to be the set of all $y \in X$ for which $y \sim x$.

Show that the set (of sets!) $\{I(x) | x \in X\}$ is a partition of X , that is,

- For all x and y , either $I(x) = I(y)$ or $I(x) \cap I(y) = \emptyset$.
- For every $x \in X$, there is $y \in X$ such that $x \in I(y)$.

Solution:

(i) Take any $x, y \in X$.

We first show that if $I(x) \cap I(y) \neq \emptyset$, then $I(x) = I(y)$. If $I(x) \cap I(y) \neq \emptyset$, then there exists $z \in I(x) \cap I(y)$ and thus $z \sim x$ and $z \sim y$. Let a be some element in one of the sets, suppose $a \in I(x)$. Thus $a \sim x$. By symmetry of the indifference relation, $z \sim x$ implies $x \sim z$. Then by transitivity of $\sim$, $x \sim y$ and thus $a \sim y$. Hence, $a \in I(y)$.

Second, if $I(x) \cap I(y) = \emptyset$, then there is no $z \in I(x) \cap I(y)$. Moreover both $I(x)$ and $I(y)$ are non-empty because of the reflexivity of $\sim$. Hence, $I(x) \neq I(y)$.

(ii) Take $x \in X$. Since $\sim$ is reflexive, $x \sim x$. Thus, $x \in I(x)$.

4. (Rubinstein, Lecture 1, Problem 3. The solution is taken from Rubinstein's solution manual.) Let Z be a finite set and let X be the set of all nonempty subsets of Z . Let $\succsim$ be a rational preference relation on X (not Z). An element $A \in X$ is interpreted as a "menu", that is, "the option to choose an alternative from the set A ".

Consider the following two properties of rational preference relations on X :

- (i) If $A \succsim B$ and C is a set disjoint to both A and B , then $A \cup C \succsim B \cup C$, and if $A \succ B$ and C is a set disjoint to both A and B , then $A \cup C \succ B \cup C$.
- (ii) If $x \in Z$ and $\{x\} \succ \{y\}$ for all $y \in A$, then $A \cup \{x\} \succ A$, and if $x \in Z$ and $\{y\} \succ \{x\}$ for all $y \in A$, then $A \succ A \cup \{x\}$.
- (a) Discuss the plausibility of the properties in the context of interpreting $\succsim$ as the attitude of the individual toward sets from which he will have to make a choice in a "second stage".

Solution:

Consider the following interpretation of the formal model: Z contains all alternatives that are potentially available for choice at the end of the decision process, and a set A is a set of candidate alternatives to be considered seriously in the second stage. If we have in mind an agent who has certainty about her second stage decision, ie. about her preference relation on Z and about the availability of candidates, then (1) is problematic:

If the best element in menu A is better than the best in menu B , and menu C includes an even better element, then $A \succ B$ but $A \cup C \sim B \cup C$, violating (1).

Also, if any element of menu A is strictly better than $z \in Z$, then $A \succ \{z\}$ but $A \sim A \cup \{z\}$, violating (2).

The properties make more sense if the decision maker has in mind a tiny possibility that she will make a mistake in her choice or that there is a possibility that a candidate will not be feasible at the end.

- (b) Provide an example of a rational preference relation that:
 - (i) satisfies the two properties.

Solution:

The relation $\succsim$ defined by $A \succsim B$ if $|A| \geq |B|$ satisfies obviously (i) and satisfies (ii) in a trivial way (since for all x and y we have $\{x\} \sim \{y\}$).

A more interesting class of examples: Let Z be divided to two sets *Good* and *Bad*. Define a preference relation by the utility function $u(A) = |A \cap \text{Good}| - |A \cap \text{Bad}|$. This satisfies both properties. (This class includes the previous one if we interpret $|A|$ as $|A \cap \text{Good}| - |A \cap \text{Bad}|$.)

(ii) satisfies the first but not the second property.

Solution:

Let $z^* \in Z$. Define $\succsim$ over X whereby $A \succ B \Leftrightarrow z^* \in A, z^* \notin B$ and $A \sim B$ otherwise.

(i) Let $A, B, C \in X$ be such that $A \succsim B$ and C is disjoint to both A and B . If $A \succ B$, then $z^* \in A$ and $z^* \notin B, C$, which implies that $A \cup C \succ B \cup C$. If $A \sim B$, then either z^* is a member of both sets $A \cup C$ and $B \cup C$ or of none. In both cases $A \cup C \sim B \cup C$.
(not ii) Let $A = \{z^*\}$ and $y \neq z^*$. Then $A \succ \{y\}$ but $A \sim A \cup \{y\}$ violating the second part of (ii).

More generally, attach to each element x a non-negative number $v(x)$ and define a preference relation by a utility function $u(A) = \sum_{a \in A} v(a)$. Then the preference relation satisfies (i) but not (ii).

(iii) satisfies the second but not the first property.

Solution:

Let $\succsim^*$ be a rational preference relation on Z . Define $\succsim$ by $A \succsim B$

if

(a) the $\succsim^*$ -best element in A is strictly better than the $\succsim^*$ -best element in B , or

(b) the agent is indifferent between the best elements but the $\succsim^*$ -worst element in A is weakly better than the $\succsim^*$ -worst element in B .

(not i) Let $a \succ^* b \succ^* c \succ^* d$. Then $\{b\} \succ \{c\}$, but $\{b\} \cup \{a, d\} \sim \{c\} \cup \{a, d\}$.

(ii) Let $A \in X$ and $z \in Z$. If z is strictly $\succsim^*$ -better (strictly $\succsim^*$ -worse) than all $a \in A$, then $A \cup \{z\} \succ A$ ($A \succ A \cup \{z\}$).

(c) Show that if there are x, y , and $z \in Z$ such that $\{x\} \succ \{y\} \succ \{z\}$, then there is no rational preference relation satisfying both properties.

Solution:

Assume $\succsim$ satisfies (i) and (ii), with $\{x\} \succ \{y\} \succ \{z\}$ for some $x, y, z \in Z$. From (ii), $\{x\} \succ \{x, y\}$ and $\{y, z\} \succ \{z\}$. Applying (i) to the above, $\{x, z\} \succ \{x, y, z\}$ and $\{x, y, z\} \succ \{x, z\}$, a contradiction.

5. (Rubinstein, Lecture 1, Problem 4. Solution from Rubinstein's solution manual) Let $\succ$ be an asymmetric binary relation on a finite set X that does not have a cycle, that is there is no finite sequence of elements $x_1, x_2, \dots, x_K$, where $K > 2$, such that $x_1 \succ x_2 \succ \dots \succ x_K \succ x_1$. Show (by induction on the size of X) that $\succ$ can be extended to a complete ordering (i.e., a complete, asymmetric, and transitive binary relation).

Solution:

Note that if a set A is finite and $\succ$ is an acyclic relation on A (there are no cycles), then there must exist an $x \in A$ such that there is no

$y \in A$ such that $y \succ x$.

Since X is finite, then there exists an $x_1 \in X$ such that there is no $y \in X$ such that $y \succ x_1$. Define $x_1 \succ^* y$ for all such $y \in X - \{x_1\}$. Again, there exists an $x_2 \in X - \{x_1\}$ such that there is no $y \in X - \{x_1\}$ such that $y \succ x_2$. Define $x_2 \succ^* y$ for all such y , and so on. By induction we can define $\succ^*$ for all $x \in X$. By construction, the relation $\succ^*$ is complete, asymmetric, extends $\succ$ and transitive: let $x_i \succ^* x_j$ and $x_j \succ^* x_h$. Then $i < j$ and $j < h$ and therefore, $x_i \succ^* x_h$.